



**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 2

Duration : 1 Hour
Total Marks: 30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I Navin Kumar (Name / Reg No) 192511015 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Navin Kumar

Backtracking Search - Technician

Problem:-

Assign three technicians (Tech 1, Tech 2, Tech 3) to three machines (x, y, z) using Backtracking search.

Constraints:-

1. Tech 1 cannot repair Machine z.
2. Tech 2 must be assigned before Tech 3 (Machine

Order : $x < y < z$)

3. Only one technician per machine
4. Tech 3 cannot repair Machine x.
5. Every technician must be assigned exactly one machine.

Backtracking steps

Step 1: Assign Tech 1

Tech 1 $\rightarrow$ x

Tech 1 $\rightarrow$ y

Tech $\rightarrow$ z (Constraint 1 violated)

choose Tech 1 $\rightarrow x$

step 2:- Assign Tech 2 Remaining machines : y, z

- Tech 2 $\rightarrow y$

- Tech 2 $\rightarrow z$

choose Tech 2 $\rightarrow y$

step 3: Assign Tech 3 Remaining machines : z

- Tech 3 $\rightarrow z$

- Tech 3 $\rightarrow x$ (constraint 4 violated)

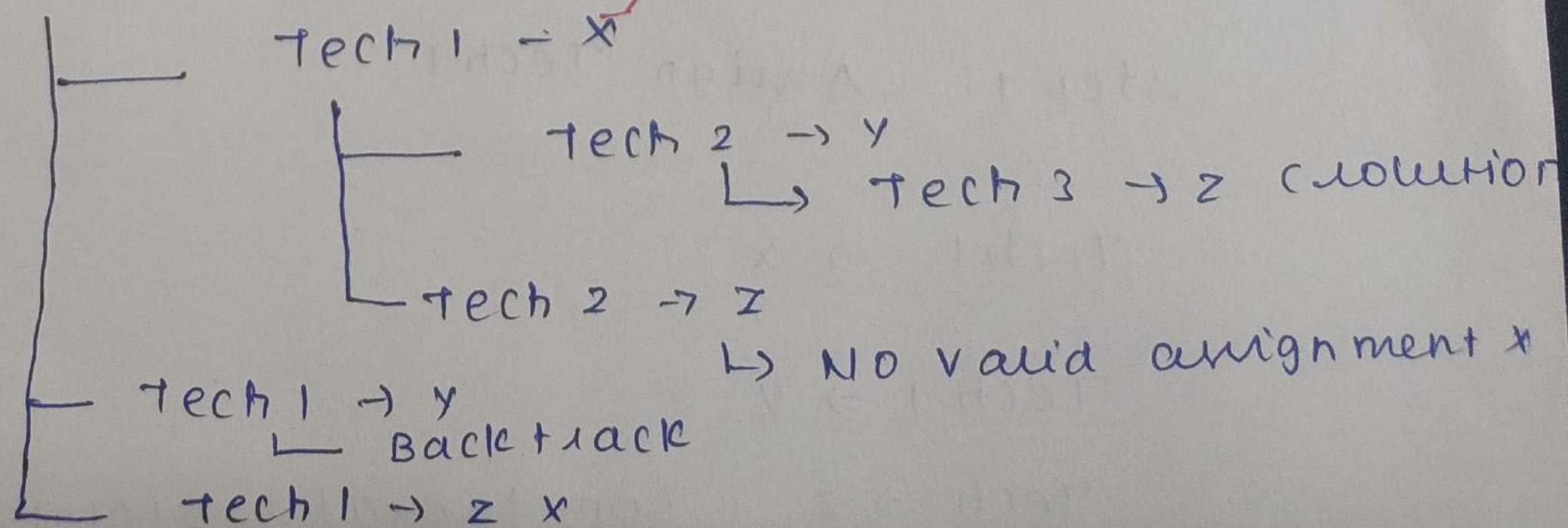
Assign Tech 3 $\rightarrow z$

Constraint Verification.

- Tech 1 is not assigned to z.
- Tech 2 is assigned before Tech 3 (y before z)
- Only one technician is assigned to each machine.
- Tech 3 is not assigned to x.
- Every technician has exactly one machine.

Search Tree:-

Start



Final Valid Schedule:-

Technician	Machine
Tech 1	Machine X
Tech 2	Machine Y
Tech 3	Machine Z

2. A search Algorithm:-

Problem:-

A warehouse robot moves in a 5×5 grid from start (S) to goal (G)

Constraints:-

- Movement up, down, left, right
- cost of each move = 1
- cannot move through obstacle cells.
- Heuristic = Manhattan distance.

A Evaluation Function

$$f(n) = g(n) + h(n)$$

where:-

- $g(n)$ = cost from start to current node.
- $h(n)$ = Manhattan distance to goal.
- $f(n)$ = Total estimated cost.

Node Expansion

Step	Node	gcn)	hcn)	fcn)
1.	S(0,0)	0	8	8
2.	C(0,1)	1	7	8
3.	C(0,2)	2	6	8
4.	C(1,2)	3	5	8
5.	C(2,2)	4	4	8
6.	C(3,2)	5	3	8
7.	C(3,3)	6	2	8
8.	C(3,4)	7	1	8
9.	u(4,4)	8	0	8

Optimal path :-

S $\rightarrow$ C(0,1) $\rightarrow$ C(0,2) $\rightarrow$ C(1,2) $\rightarrow$ C(2,2)
 $\rightarrow$ C(3,2) $\rightarrow$ C(3,3) $\rightarrow$ C(3,4) $\rightarrow$ u

Total cost

• Number of moves = 8

• Optimal path cost = 8

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

$22 + 21 + 20 + 15 + 10$
 $= 88$
 $\frac{88}{5} = 17.6$